

Appendix 1.2: TSB BANK TOKENOMICS AND TWO-CIRCUIT FINANCIAL-CLEARING REGULATIONS

Institutional Executive Summary & Regulatory Scope
This specification defines the official tokenomic framework, financial-clearing architecture, and monetary regulations for Transceptor Sovereign Bank (TSB) within the Transceptor Global ecosystem. Operating under Swiss DLT-Act and Singapore MAS regulatory standards, this document establishes the operational parameters for two-circuit monetary clearing (T_{nat} and T_{net}), gross entry/exit fee splitting, reserve capital preservation, internal turnover fees, FX convergence mechanics, and the canonical protocol distribution waterfall.

1. Circuit Architecture & Gross In-Split

The financial infrastructure of Transceptor Global is split into two algorithmically isolated and independently regulated clearing circuits to safeguard local sovereign currencies while providing a global energy-backed digital clearing unit.

1.1. Circuit A: Sovereign / Local Currency Circuit (T_{nat})

Circuit A handles interactions with traditional sovereign fiat currencies and national banking infrastructure (T_{nat}). It operates through Joint Venture Special Purpose Vehicles (SPV 51/49) established with licensed domestic banking institutions.

Capital entering Circuit A triggers an automated gross in-split fee of 2.50% at the clearing gateway prior to minting T_{nat}:

- **Protocol Share (2.00% gross):** Transferred directly to the canonical protocol allocation waterfall for ecosystem regeneration, compute infrastructure, and reserve pools.

- **Bank / Clearing Share (0.50% gross):** Retained by TSB and domestic banking partners to cover fiat processing overhead, regulatory compliance (ZK-KYC/AML), and custodial OpEx.

1.2. Circuit B: Global Energy Voucher Circuit (T_{net})

Circuit B governs T_{net}, the primary global settlement voucher of Transceptor Global. Each T_{net} token is algorithmically linked to 1 kWh of verified surplus clean energy under the Anthropocene Sun Constant standard.

Capital entering Circuit B or converting into T_{net} incurs a gross in-split fee of 3.50%:

- **Protocol Share (3.00% gross):** Directed to the canonical protocol waterfall to support global compute clusters, thermodynamic regeneration buffers ($S_{reserve}$), and cooperative development pools.
- **Bank / Clearing Share (0.50% gross):** Allocated to TSB to cover hardware enclave isolation (TEE), cross-border clearing rails, and Real World Asset (RWA) custody management.

Circuit Identifier	Primary Asset Class	Total Gross Fee
Circuit A	T_{nat} (Sovereign Fiat)	2.50%
Circuit B	T_{net} (Energy Voucher)	3.50%

2. Target Deposit Lock, Capital Immunity & 100% RWA Deposition

To ensure systemic solvency and eliminate fractional-reserve risks, TSB enforces strict capital backing rules for all minted digital assets.

2.1. Target Deposit Lock ($20 Soulbound Reserve)

Upon onboarding, each participant holding a verified W3C Decentralized Identifier (DID) establishes a non-transferable $20 Soulbound reserve deposit:

- **Principal Protection:** 100% of the $20 deposit is preserved in full and deposited into sovereign short-term paper.
- **Service Fee Deduction:** The standard annual account maintenance fee ($20/year) is charged strictly post-facto on day 365, ensuring zero initial capital erosion.

2.2. Capital Immunity Threshold ($1,000+ Lock)

Participants who maintain a capital deposit lock of $1,000 or greater for a minimum continuous period of 365 days receive complete **Capital Immunity**:

- **Fee Exemption:** Annual account maintenance and standard administrative service fees are completely waived (reduced to $0).
- **Yield Substitution:** Coupon income generated by the underlying RWA portfolio offsets operational expenses on behalf of the participant.

2.3. 100% RWA Deposition Standard

All reserve funds backing T_{nat} and T_{net} are 100% deposited into high-grade, liquid Real World Assets (RWA)—specifically 1 to 3-year sovereign Treasury Bills (e.g., US T-Bills, Swiss Confederation Bonds) with full principal protection. Unbacked token issuance or algorithmic uncollateralized lending is prohibited in the core microcode.

3. Internal P2P Turnover Fee

To discourage speculative velocity churn while fostering authentic peer-to-peer economic activity within the Decentralized Agora, all internal transfers, P2P barter exchanges, and asset transactions incur a baseline 1.50% turnover fee.

3.1. Fee Allocation Breakdown

The 1.50% turnover fee captured by the smart-contract clearing module (`clearing_gateway.hpp`) is distributed according to the following fixed proportions:

- **50% — Infrastructure Fund (F_{infra}):** Funds GPU/NPU compute cluster scaling, TEE enclave maintenance, and network routing nodes.
- **30% — Cooperative Pool & MPK OpEx (P_{coop}):** Allocates funds to local community initiatives, regional Oikoumene development, and Meritocratic Competence Passport operational overhead.
- **20% — Founders IP Level 3:** Remitted to core technology licensors for foundational system architecture, continuous research, and protocol upgrades.

4. Exit Policy, Secondary P2P Assignment & Early Withdrawal Penalty

TSB enforces clear rules governing liquidity exit and secondary asset transfers to protect systemic reserve stability.

4.1. Standard Exit Redemption

Participants may redeem T_{nat} or T_{net} for underlying fiat currency upon reaching the maturity date of their underlying RWA deposit locks (12 to 36 months). Outbound redemptions are subject to standard gross exit splits corresponding to their respective circuit (2.50% for Circuit A, 3.50% for Circuit B).

4.2. Secondary P2P Market Assignment

Participants seeking liquidity prior to deposit maturity are encouraged to utilize the secondary P2P marketplace within the Decentralized Agora. Locked capital positions or T_{net} vouchers may be assigned to secondary peer buyers via atomic smart contracts without liquidating the underlying sovereign T-Bill portfolio, incurring only the standard 1.50% P2P turnover fee.

4.3. Early Withdrawal Penalty

In cases where emergency early fiat redemption is requested directly from TSB prior to the 365-day threshold, an automated early withdrawal protocol is triggered:

- **Penalty Fee:** A flat 5.00% early termination fee is assessed on the principal balance.
- **Yield Forfeiture:** All accrued interest and coupon earnings are forfeited and transferred directly into the ecosystem's Regeneration Reserve Fund ($S_{reserve}$).

5. FX Spread Mechanics & Entropy Parity

Cross-circuit monetary exchange between T_{nat} and T_{net} is regulated through automated liquidity market-maker (ALMM) modules inside TEE hardware enclaves.

5.1. Instant Spread Capture ($T_{nat} \leftrightarrow T_{net}$)

When value moves between local sovereign currency (T_{nat}) and global energy vouchers (T_{net}), the clearing engine captures the instantaneous foreign exchange spread. This spread capture neutralizes latency arbitrage and prevents cross-border capital extraction by high-frequency trading bots.

5.2. Dynamic Entropy Parity Convergence

The equilibrium exchange rate between T_{nat} and T_{net} dynamically adjusts based on the regional Entropy Parity Index ($K_{entropy}$):

- **Thermodynamic Appreciation:** Regions that increase net thermodynamic efficiency (verifiable reduction of environmental and operational entropy) trigger an appreciation of T_{net} relative to T_{nat}.
- **Systemic Convergence:** This dynamic mechanism incentivizes local economies to adopt clean energy generation and sustainable resource management, aligning economic value with physical efficiency.

6. Canonical Waterfall Breakdown of Protocol Share

The protocol share retained from gross entry fees (2.00% in Circuit A and 3.00% in Circuit B) is allocated according to the **Canonical Protocol Waterfall**.

6.1. Distribution Schedule

Component	Share %	Target Fund
Infrastructure Fund	30.00%	F_{infra}
Regeneration Reserve	20.00%	$S_{reserve}$
Bank SPV Reserve	20.00%	SPV 51/49
Cooperative Pool	20.00%	P_{coop}
Founders IP Level 3	10.00%	IP Core

Automated Execution Directive
All waterfall disbursements are executed programmatically via smart contracts inside TEE hardware enclaves under oversight by the Trust-Policy-Engine (Ψ_{TPE}). Transactions

are verified using Zero-Knowledge proofs (ZK-SNARKs) to guarantee zero administrative latency, absolute transparency, and total immunity against manual administrative tampering.